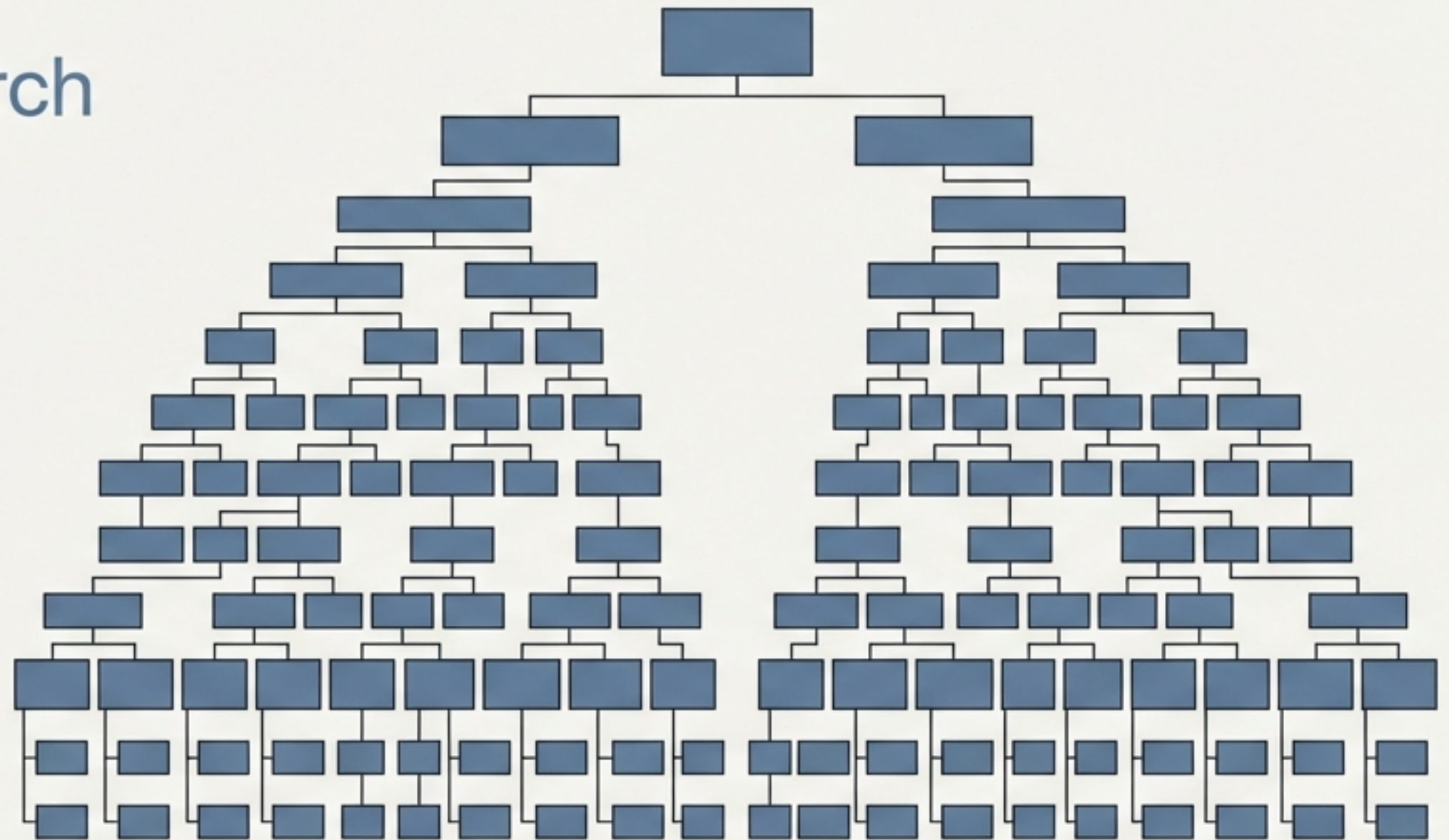


The Architecture of Hierarchical Knowledge Navigation

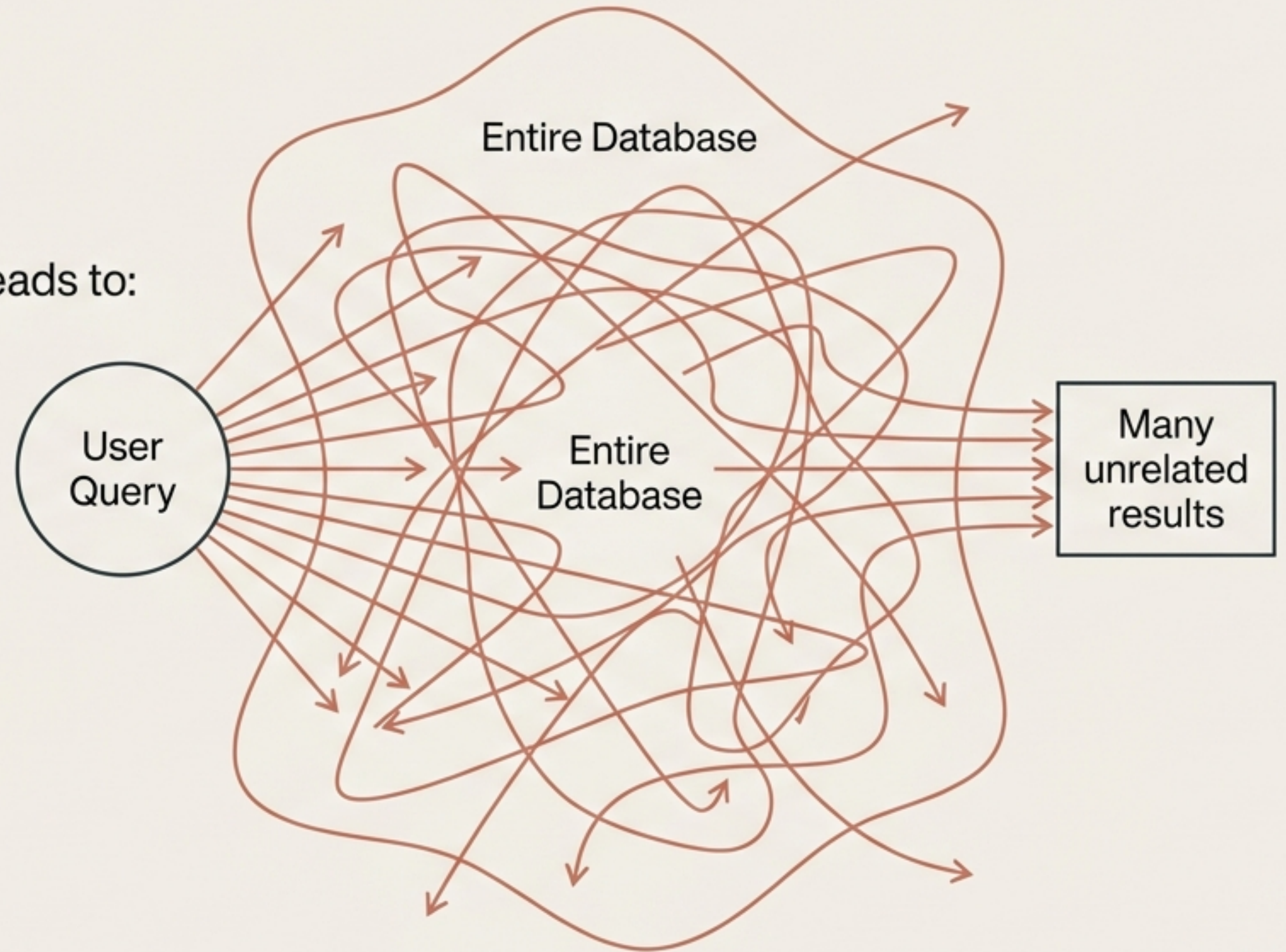
Moving beyond global search with PageIndex



The limits of global search in LLM retrieval

Large Language Models generate fluent answers, but they fail when retrieving information from an unconstrained knowledge base. Traditional global search leads to:

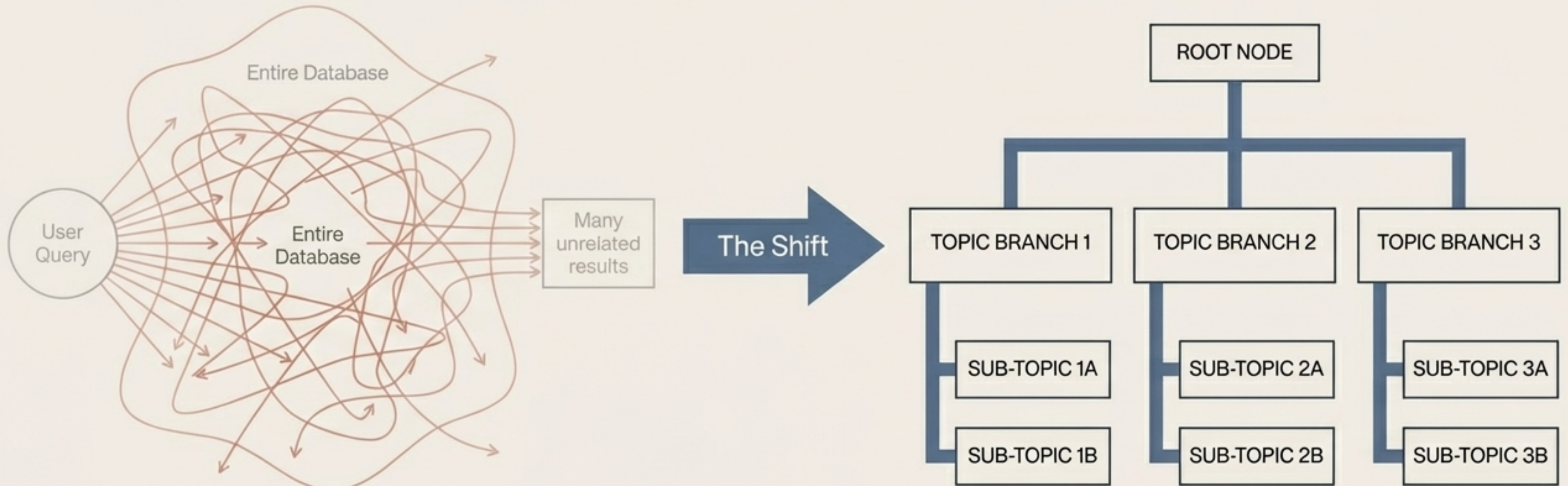
- Retrieval from unrelated topics
- Retrieval from unrelated topics
- Loss of conversational context
- Inconsistent follow-up responses



Structured knowledge navigation

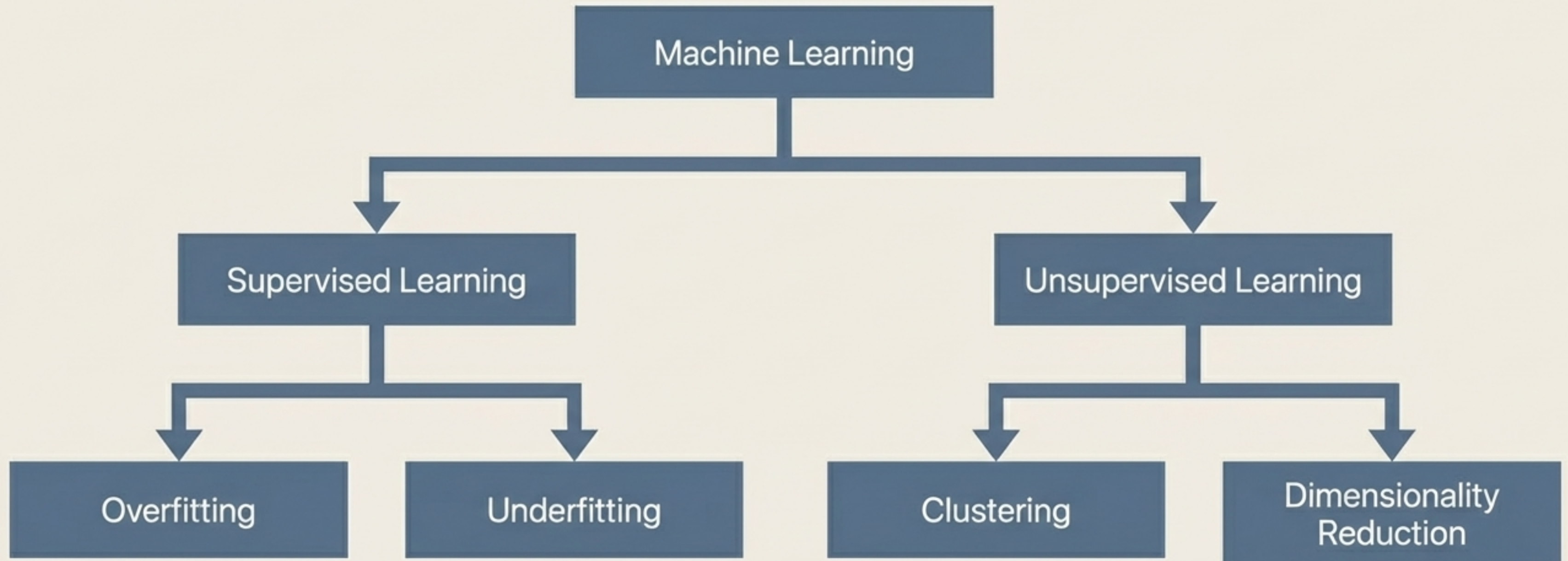
To improve reliability, AI systems require PageIndex—a structured knowledge navigation approach.

Instead of searching the entire database globally, the system navigates through an organised hierarchical tree to reach the most relevant topic. The AI is constrained to the correct knowledge branch, drastically improving accuracy and context awareness.



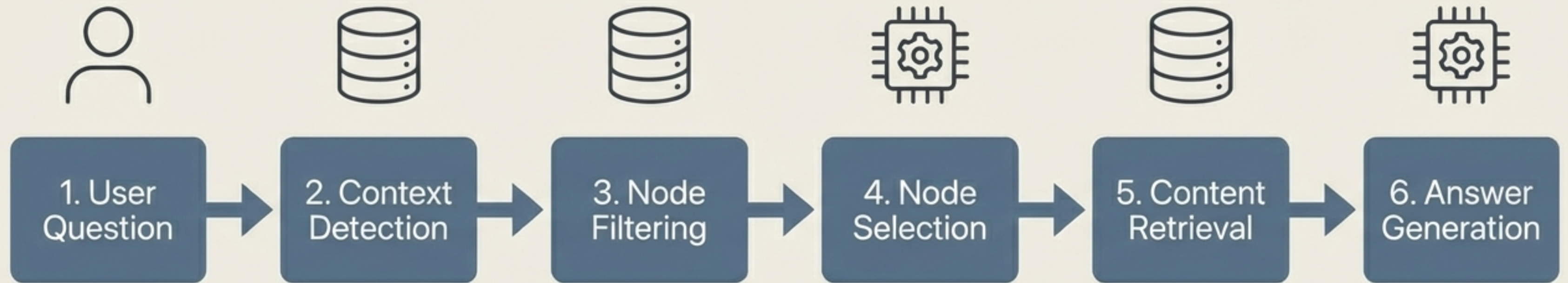
Organising knowledge into a hierarchical tree

Knowledge is categorised into distinct root topics, branches, and nodes, creating a precise navigational map for the LLM.



The six-stage navigation pipeline

PageIndex retrieves information through guided navigation. The process follows a strict linear flow from the initial query to the final generated response.



Establishing intent and position

Step 1: User Question

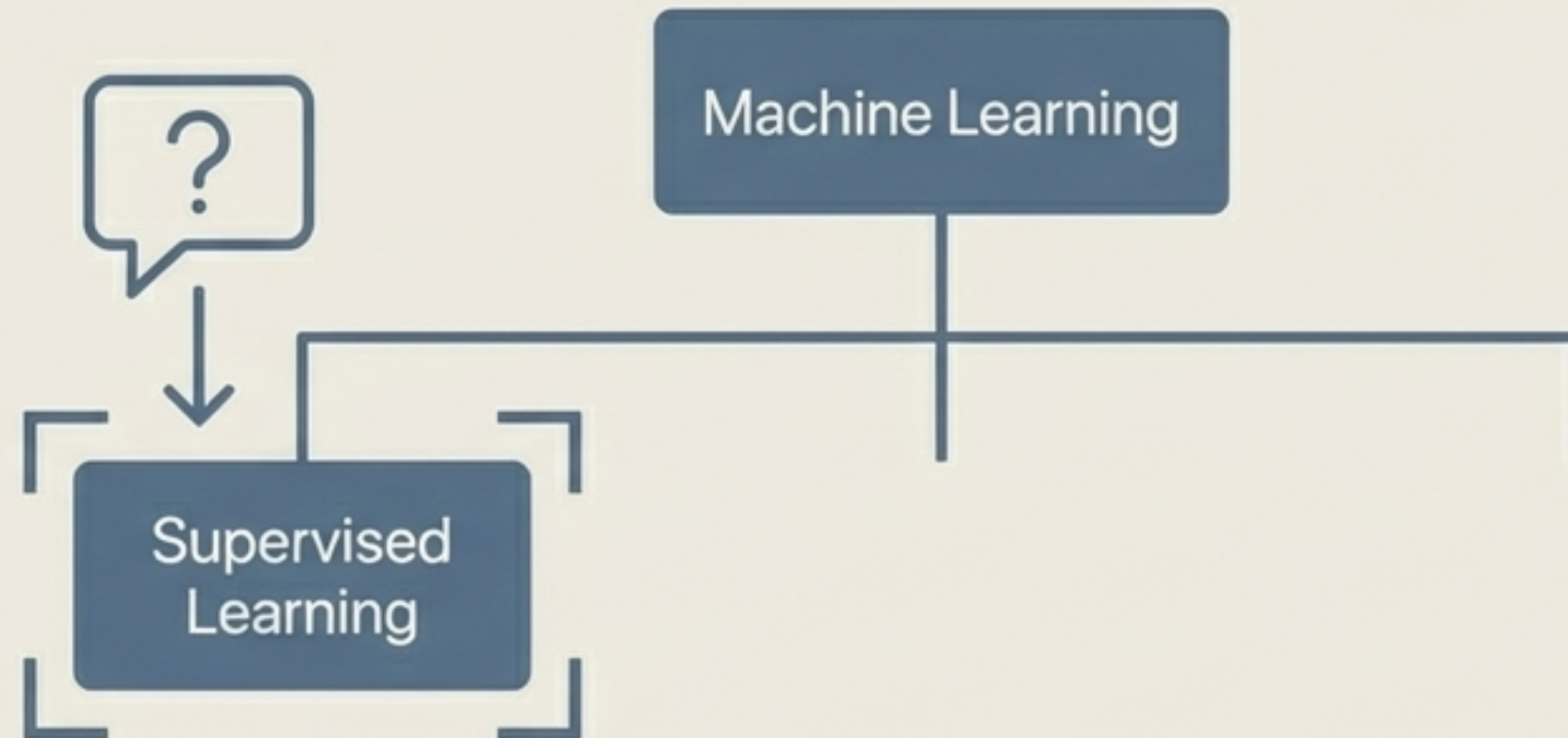
The trigger input.

Example: "Why does overfitting happen?"

Step 2: Context Detection

The system identifies the user's current position within the knowledge tree before executing any search.

Example: Current Node = Supervised Learning



Isolating the precise subtopic

Step 3: Node Filtering

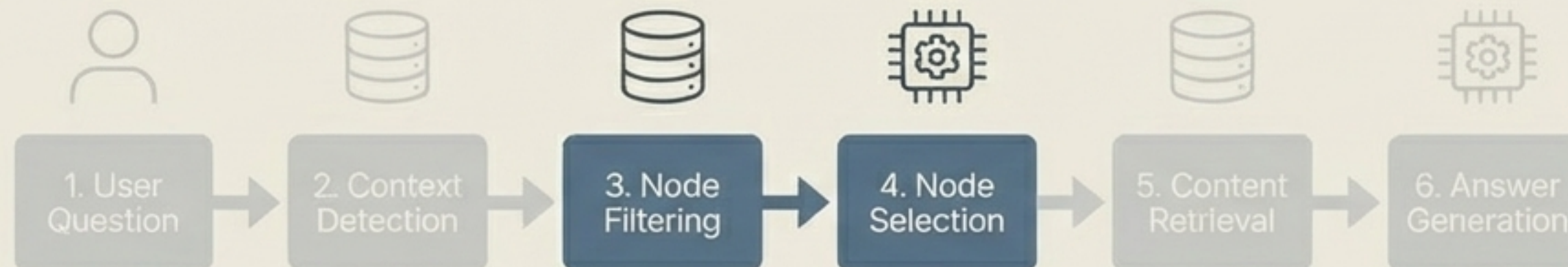
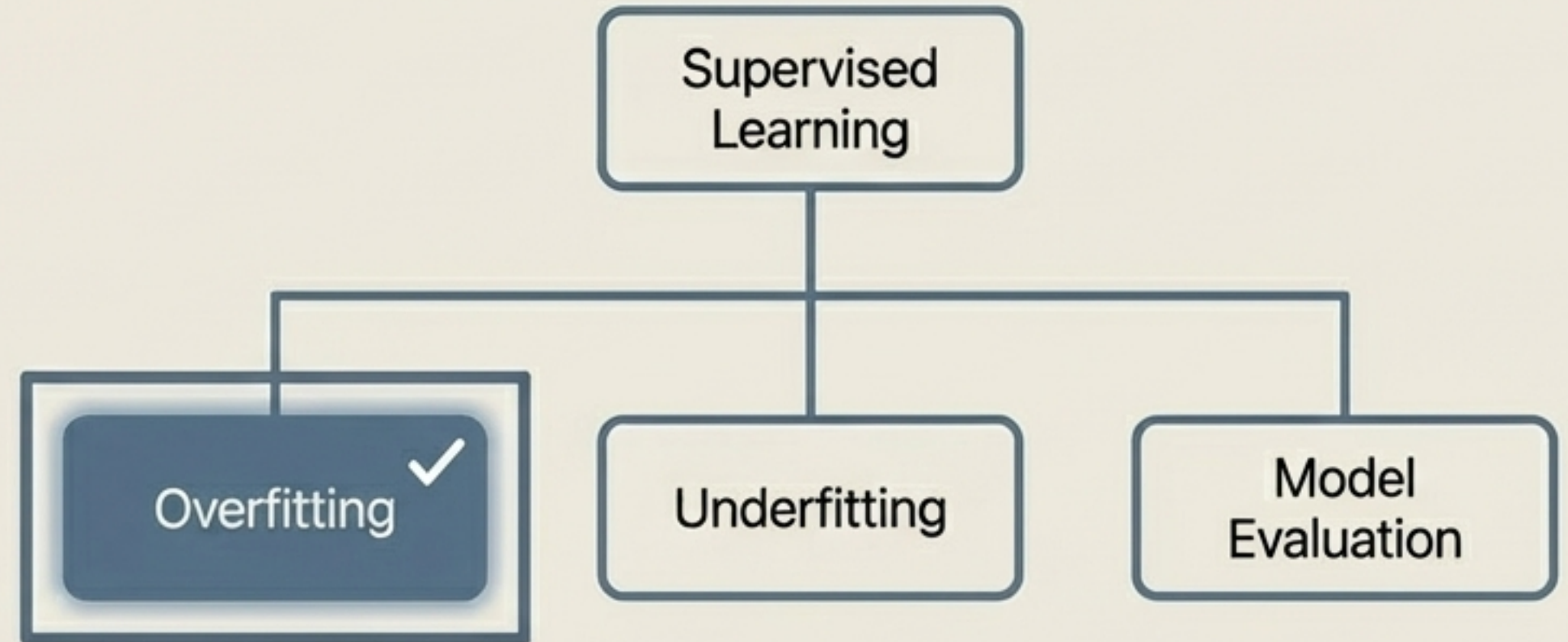
The backend retrieves only the child nodes directly related to the current topic.

Example Nodes: Overfitting, Underfitting, Model Evaluation

Step 4: Node Selection

The LLM analyses the question against this restricted list and selects the most relevant node.

Selected Node: Overfitting



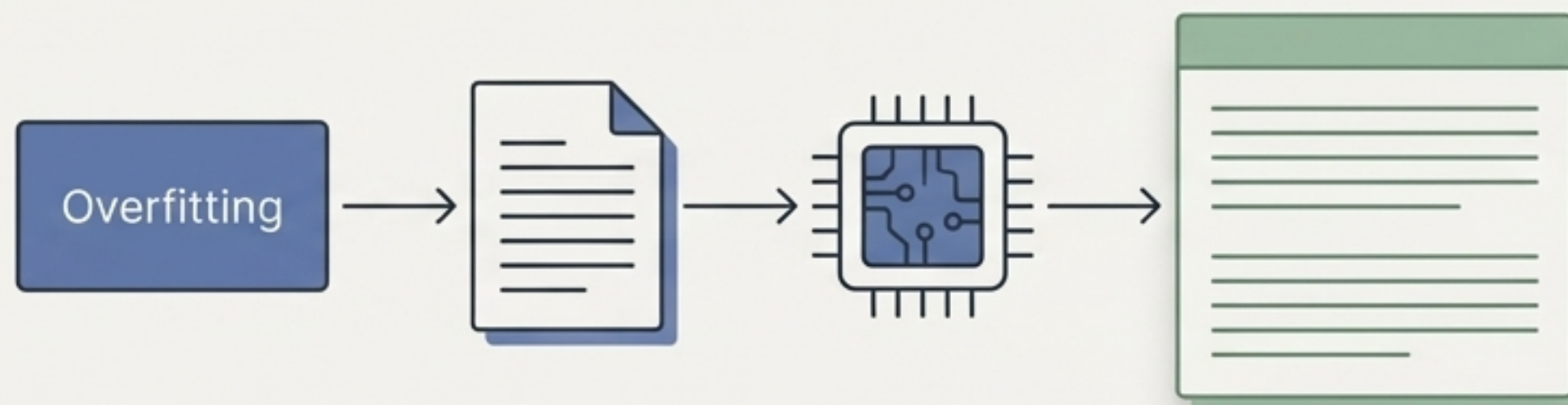
Content retrieval and answer generation

Step 5: Content Retrieval

The system pulls the specific text, data, and context associated exclusively with the selected node.

Step 6: Answer Generation

The LLM generates the final explanation, strictly using the retrieved, verified knowledge.



Constraining the search space minimises hallucination

Strong Structural Control

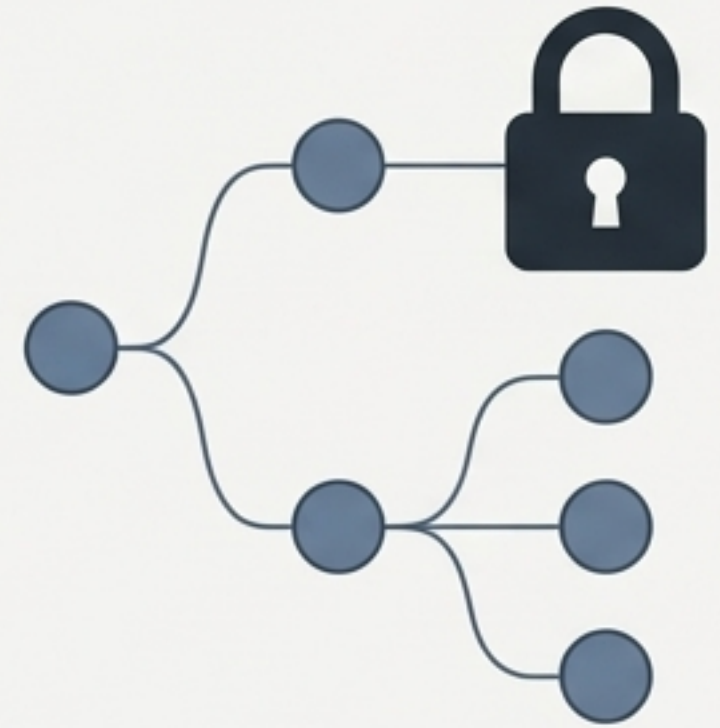
Hierarchical organisation strictly prevents the LLM from pulling data from unrelated topics.

Reduced Hallucination Risk

Responses are generated using only verified node content, drastically improving factual reliability.

Explainable Retrieval

Every generated answer traces back to a specific node in the knowledge tree, making the system entirely transparent and highly debuggable.



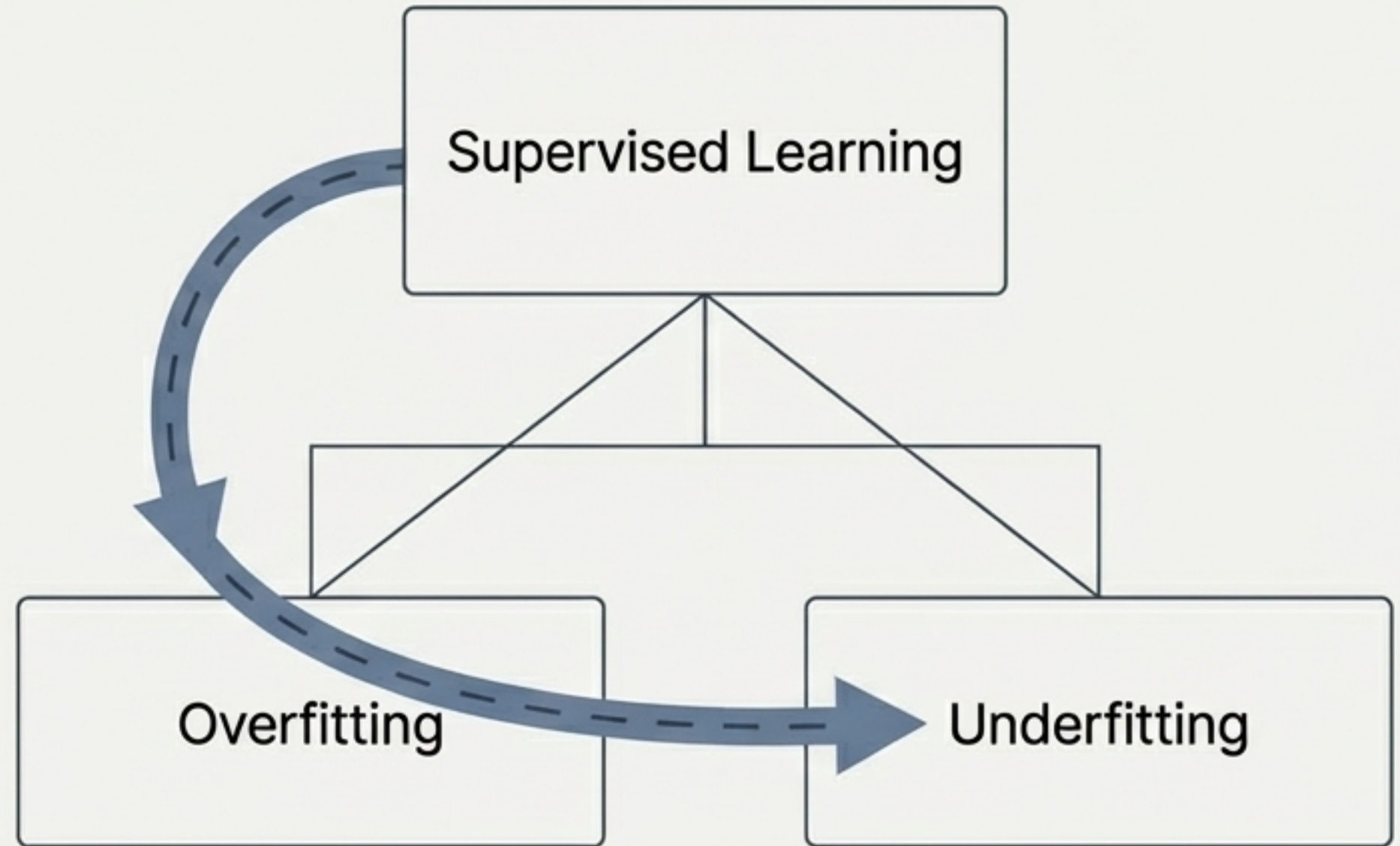
Maintaining context across follow-up queries

Context-Aware Conversations

Because the AI knows its location in the hierarchy, follow-up questions naturally move to sibling nodes within the same branch without losing the conversational thread.

Example Conversation Path:

Supervised Learning → Overfitting
→ Underfitting

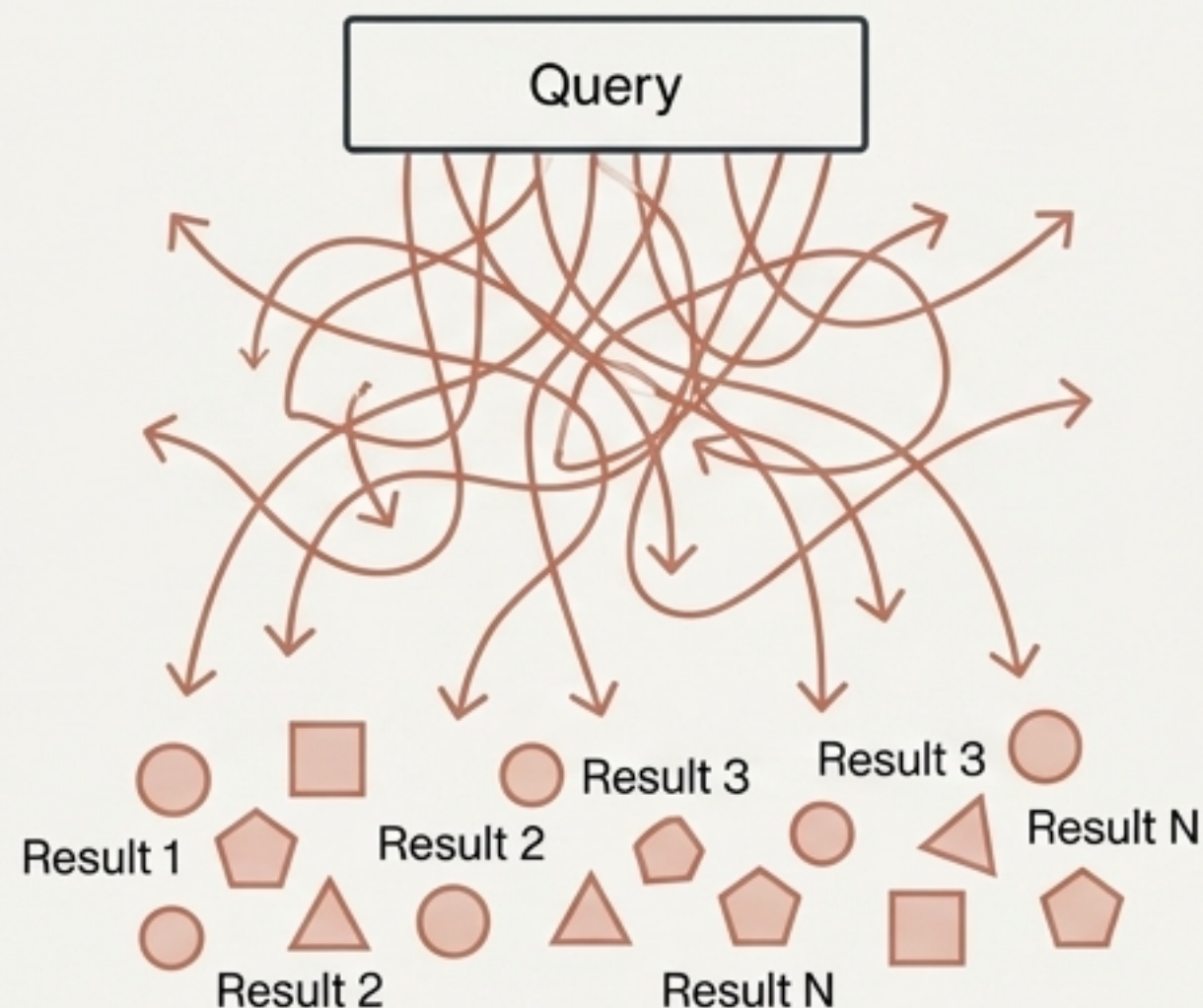


The paradigm shift: Global search vs. PageIndex

PageIndex transforms knowledge retrieval from unpredictable global search into precise, structured navigation—enabling accurate, contextual, and explainable AI.

Global Search

Query → Entire Database → Many unrelated results



PageIndex Navigation

Query → Correct Topic → Correct Subtopic → Relevant Answer

